From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2014

FULL TEST – V

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 252**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two section: **Section-A & Section-C**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) Section-A (01 – 08) contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 – 13) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

Section-A (14 – 18) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) Section-C (01 – 10) contains 10 Numerical based questions with answers as numerical value and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

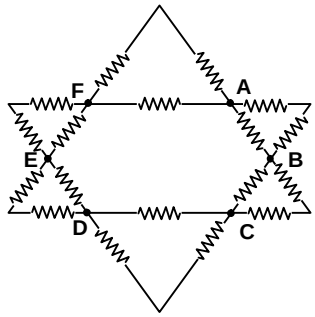
PART – I

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Resistance of each resistor is R. Then the equivalent resistance across A and B is



(A) $\frac{5}{9}R$ (B) $\frac{4}{9}R$

(C) $\frac{R}{9}$ (D) $\frac{7}{9}R$

2. Consider two coherent monochromatic (wavelength λ) sources S_1 and S_2 separated by distance d. The ratio of intensity of S_1 and that of S_2 (which is responsible for interference at point P, where detector is located) is 4. The distance of point P



from S_1 is (if the resultant intensity at point P is equal to $\frac{9}{4}$ times of intensity of S_1)

[Given $\angle S_2 S_1 P$ is 90°] (d > 0, n is a positive integer)

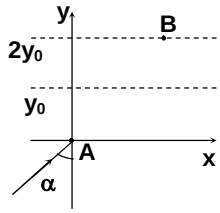
(A) $\frac{d^2 - n^2\lambda^2}{2n\lambda}$ (B) $\frac{d^2 + n^2\lambda^2}{2n\lambda}$

(C) $\frac{n\lambda d}{\sqrt{d^2 - n^2\lambda^2}}$ (D) $\frac{2n\lambda d}{\sqrt{d^2 - n^2\lambda^2}}$

Space for rough work

3. The refractive index of space changes with y, whose function is given as

$$\mu(y) = \begin{cases} 1 & \text{if} \quad y \le 0 \\ ky^2 & \text{if} \quad 0 \le y \le 2y_0 \\ \dfrac{\ell}{y^2} & \text{if} \quad 2y_0 \le y \le 4y_0 \end{cases}$$



where k and ℓ are positive constant. When a light ray is incident at point A as shown in figure. When the ray reaches to point B, it becomes parallel to xz-plane. Choose the correct statement.

(A) α is independent of k (B) α is independent of ℓ

(C) α is independent of both k and ℓ (D) none of these

4. The space has electromagnetic field which varies with time whose variation is given as

$$\vec{B} = \begin{cases} -B_0\hat{k} & \text{if} \quad 0 \le t \le \dfrac{\pi m}{qB_0} \\ -B_0\hat{j} & \text{if} \quad \dfrac{\pi m}{qB_0} \le t \le \dfrac{2\pi m}{qB_0} \\ 0 & \text{if} \quad \dfrac{2\pi m}{qB_0} \le t < \infty \end{cases} \text{ and } \vec{E} = \begin{cases} 0 & \text{if} \quad 0 \le t \le \dfrac{2\pi m}{qB_0} \\ -E_0\hat{k} & \text{if} \quad \dfrac{2\pi m}{qB_0} \le t < \infty \end{cases}$$

A charge particle having mass m and positive charge q is given velocity $v_0\hat{i}$ at origin at t = 0 sec. The coordinate of point on xy plane when it is passing through xy plane again is

(A) $\left(\dfrac{2mv_0}{q}\sqrt{\dfrac{v_0}{E_0B_0}}, \dfrac{2mv_0}{qB_0}, 0\right)$ (B) $\left(\dfrac{2mv_0}{q}\sqrt{\dfrac{v_0}{E_0B_0}}, \dfrac{mv_0}{qB_0}, 0\right)$

(C) $\left(\dfrac{mv_0}{q}\sqrt{\dfrac{v_0}{E_0B_0}}, \dfrac{2mv_0}{qB_0}, 0\right)$ (D) $\left(\dfrac{mv_0}{q}\sqrt{\dfrac{v_0}{E_0B_0}}, \dfrac{mv_0}{qB_0}, 0\right)$

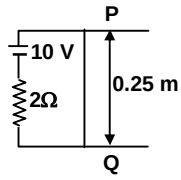
5. If the sound heard by observer, whose equation is given as
$y = 8\sin 10\pi t \cos 200\pi t$ at $x = 0$
The number of beat frequency heard by observer is 2k, then the value of k is
(A) 10 (B) 5
(C) 4 (D) none

6. Two sound waves of frequencies 100 Hz and 102 Hz and having same amplitude 'A' are interfering. A detector which can detect waves of amplitude greater than or equal to A is kept at rest. In a time interval of 12 seconds, find the total duration in which detector is active.
(A) 2sec (B) 4sec
(C) 6sec (D) 8sec

Space for rough work

7. Two identical photons of energy E ($4eV \le E \le 8eV$) fall on two different metals whose work functions are in the ratio of 1 : 2. The ratio of the kinetic energies of the most energetic electrons coming from each metal is 2 : 1. Work function of one of the metal is 4 eV. What is the energy of photon in eV.
(A) 6eV (B) 8 eV
(C) 4 eV (D) 5 eV

8. A metal wire PQ slides on parallel metallic rails having separation 0.25 m, each having negligible resistance. There is a 2 Ω resistor and 10 V battery as shown in figure. There is a uniform magnetic field directed into the plane of the paper of magnitude 0.5T. A force of 0.5 N to the left is required to keep the slide wire PQ moving with constant speed to the right. With what speed is the wire PQ moving?



(A) 14 m/s (B) 16 m/s
(C) 18 m/s (D) 20 m/s

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. In the shown figure (1) and (2), capacitors are in the steady state. Charging batteries are removed and switches S_1 and S_2 are closed at time t = 0. The plot of $\ell n I$ (I is the current in the resistor) in against time t in the resistors R_1 and R_2 are shown by the graphs 1 and 2 respectively in the figure (3). Choose the correct option(s).
(A) If $\varepsilon_1 = 2\varepsilon_2$, R_1 must be equal to $2R_2$ (B) If $R_1 = R_2$, C_1 must be less than C_2
(C) If $C_1 = C_2$, R_1 must be less than R_2 (D) R_1C_1 is equal to R_2C_2

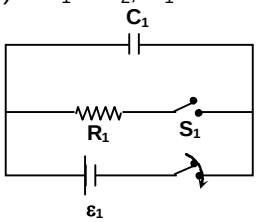


figure (1)

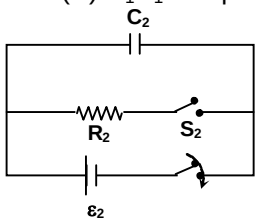


figure (2)

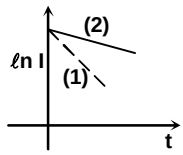
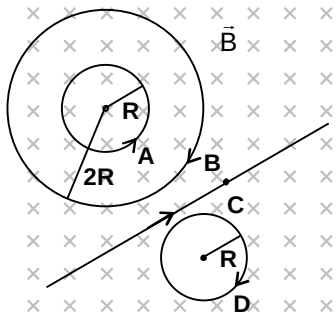


figure (3)

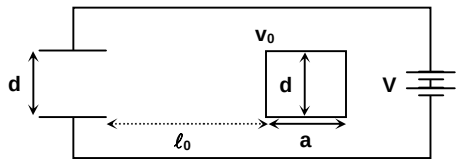
Space for rough work

10. Four particles A, B, C and D of masses m_A, m_B, m_C and m_D respectively, follow the paths shown in the figure, in a uniform magnetic field. Each particle moving with same speed. Q_A, Q_B, Q_C and Q_D are the specific charge of particles A, B, C and D respectively (assume that the motion of each particle is in the same plane perpendicular to the magnetic field).



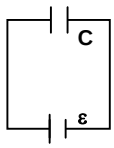
(A) $Q_A < Q_B < Q_C < Q_D$
(B) $Q_D < Q_B < Q_C < Q_A$
(C) charge on the particle B and particle D is of same nature
(D) Work done by magnetic force on the particle C is minimum as compared to other particle

11. A parallel plates capacitor is made of square conducting plates of side a and the separation between plates is d. The capacitor is connected with battery of emf V volt as shown in the figure. There is a dielectric slab of dimension a × a × d with dielectric constant k. At t = 0, dielectric slab is given velocity v_0 towards capacitor as shown in the figure. (Neglect the effect of gravity and electrostatic force acting on the dielectric when dielectric is out side of capacitor. Also ignore any type frictional force acting on the dielectric during its motion) let the x be the length dielectric inside the capacitor at t = t sec. [$\ell_0 \gg a$]



(A) Motion of dielectric slab is periodic but not simple harmonic motion
(B) Motion of dielectric slab is simple harmonic motion
(C) At any time, the slope of graph of total energy verses x is twice the slope of graph of potential energy verses x.
(D) The value of maximum energy stored in the system is $\frac{1}{2}mv_0^2 + \frac{\varepsilon_0 a^2 V^2}{2d}(2k-1)$

12. A capacitor of capacitance 'C' is connected to a battery of emf ε as shown. After full charging a dielectric of same size as that of capacitor & dielectric constant k is inserted. Then choose the **INCORRECT** statement(s).



(Capacitor remains always connected to battery)
(A) electric field between plates of capacitor remains same
(B) charge on capacitor increases to 2kC ε
(C) energy stored in capacitor decreases
(D) electric field between plates of capacitor increases

Space for rough work

13. The equation of a process of diatomic gas is $P^2 = \alpha^2 V$ where α is a positive constant. Then choose the **INCORRECT** option(s).

(A) Work done by gas for a temperature change T is $\frac{2}{3}\alpha nRT$

(B) The change in internal energy is $\frac{5}{2}nRT$ for a temperature change T

(C) The specific heat for the process is $\frac{19}{9}R$

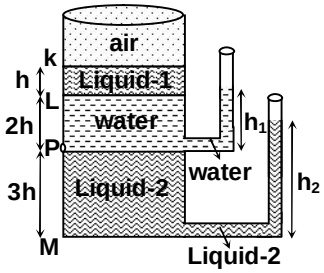
(D) The change in internal energy for a temperature change T is $\frac{5}{2}\alpha nRT$

Comprehension Type

This section contains **2 paragraphs**. Based upon one of paragraph **2 multiple choice questions** and based on the other paragraphs **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of **which ONLY ONE** is correct.

Paragraph for Question Nos. 14 and 15

A container contains three liquid as shown in the figure. The specific gravity of liquid I and liquid II are 0.8 and 1.2 respectively. The is small hole having area 10^{-6} m^2 at point P as shown in the figure. Initially hole is closed. The cross-sectional area of container is 50 cm^2. Assume all liquids are ideal and neglect any type of friction between liquid and container. Read the passage carefully and answer the following question(s) (given that g = 9.8 m/s^2 and h = 28 cm)



14. The value of h_2/h_1 is

(A) 9/5 (B) 20/7

(C) 40/21 (D) 40/7

15. At some instant, hole is open, the speed of water coming out from hole just after the opening of hole is

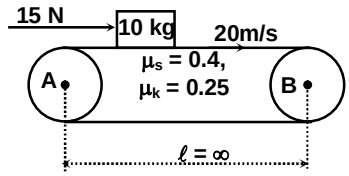
(A) 7.84 m/s (B) 3.92 m/s

(C) 1.96 m/s (D) none of these

Space for rough work

Paragraph for Question Nos. 16 and 18

A block of mass 10 kg is put gently on a belt-conveyor system of infinite length at t = 0 sec, which is moving with constant speed 20 m/sec rightward at all time, irrespectively of any situation by means of a motor-system as shown in the figure.
tant force of magnitude 15 N is applied on the block continuously during its motion.



16. The nature of friction force acting on the block of mass 10 kg during its motion is
(A) kinetic only (B) Static only
(C) some time kinetic some time static (D) can't be predicted

17. Work done by the kinetic friction on the block of mass 10 kg is
(A) 1250 Joule (B) 2500 Joule
(C) −1250 Joule (D) none

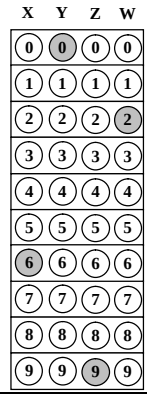
18. The magnitude of acceleration of the block of mass 10 kg at t = 6 sec is
(A) 4 m/s^2 (B) 3 m/s^2
(C) 2 m/s^2 (D) none

Space for rough work

SECTION – C

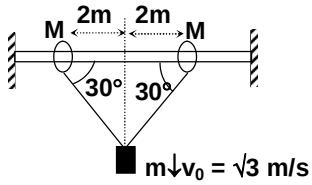
Integer Answer Type

This section contains **10 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



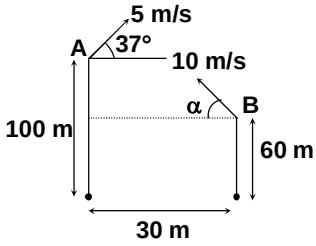
1. The xz plane is the boundary between two transparent medium. Medium –I with $y \geq 0$ has refractive index μ and medium –II with $y \leq 0$ has a refractive index 1. A ray of light in medium-I, given by vector $\vec{n} = \hat{i} - \sqrt{3}\hat{j}$ is incident on the plane of separation. If the reflected and refracted rays make an angle of 90° with each other, then the value of μ is $\sqrt{k}$. Find the value of k.

2. Two rings, each of mass M = 100 gm are constrained to move along a fixed horizontal rod. An ideal string is connected with rings and a small m = 200 gm is connected to the mid point of string. At any moment m is moving downward with velocity $\sqrt{3}$ m/s. The speed of ring of M at the same moment is



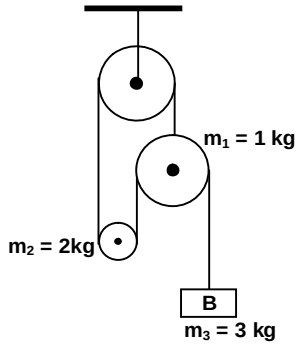
3. Pressure variation due a sonic wave propagating along positive x-direction is given by the equation $\Delta P(x) = \Delta P_0 \sin[2\pi(vt - x)]$. One end of an open organ pipe is closed by a cap and held at rest with its axis parallel to x-axis and the open end at the origin. What should be the minimum value of frictional force between the cap and the pipe so that pipe will remain closed. Given $\Delta P_0 = 2 \times 10^3$ N/m^2 and cross sectional area of the tube is equal to 5×10^{-4} m^2,

4. Two particle are projected simultaneously at point A and point B from two towers as shown in the figure. If they collide then the value of α is (a $\times$ 10 + b) in degree. Find the value of (a – b). (Neglect any type of frictional force acting on the particles during motion. (take g = 10 m/s^2)(where a and b are positive integer, a < 10 and b < 10)

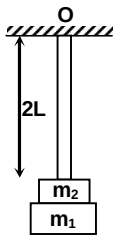


Space for rough work

5. In the given system which mass (in kg) will move with acceleration greater than acceleration due to gravity(g).



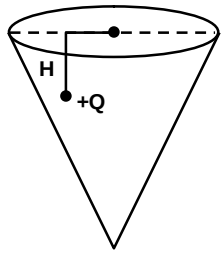
6. A string of negligible mass, natural length L has Young's modulus Y. The string hangs from roof with masses m_1 and m_2 as shown in the figure. If mass m_2 is removed gently, the mass m_1 is just able to bounce back upto point O. Find the ratio m_2/m_1. (string will not obstruct the motion of mass m_1 and system is initially in equilibrium).



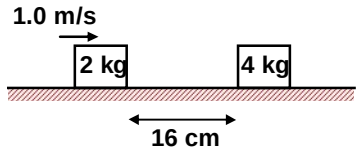
7. Use the following equation $(m)^{ft}(a-c)^2 = x\left[d - kx - \frac{mc}{t}\right]$; where k = spring constant, x = distance, m = mass, t = sec and a, c, d and f are some other physical quantities. If the dimensional formula of f is $[M^{\ell}L^{m}T^{-n}]$, then the value of $\ell + m + n$ is

Space for rough work

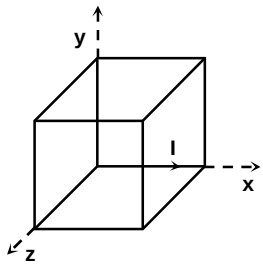
8. A charge +Q is located somewhere inside a vertical cone such that the depth of the charge from the free surface of the cone is H. It is found that the flux associated with the cone with the curved surface is $\frac{3Q}{5\varepsilon_0}$. If the charge is raised vertically through a height 2H, the flux associated with the cone with the curved surface is $\frac{\eta Q}{20\varepsilon_0}$. What should be the value of η?



9. The friction coefficient between the horizontal surface and each of the block shown in the figure is 0.2. The collision between the blocks is perfectly elastic. Find the separation (in cm) between them when they come to rest.
Take g = 10m/s^2.



10. A cubical frame having edge length a, each having a resistance R is placed in a gravity free space such that axes are along sides of cube. Now a cylindrical region of time varying magnetic field is created around the cube having axis in direction of vector $\left(\hat{i}+\hat{j}+\hat{k}\right)$. It was found that current through edge along x axis is I = $\left|\eta\frac{a^2}{R}\frac{dB}{dt}\right|$, where B is the instantaneous magnetic field. What should be the value of η?



Space for rough work

Chemistry

PART – II

SECTION – A

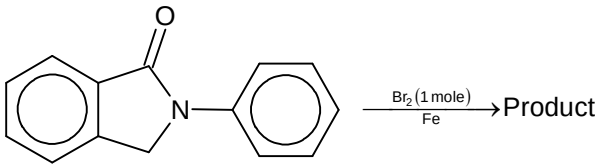
Straight Objective Type

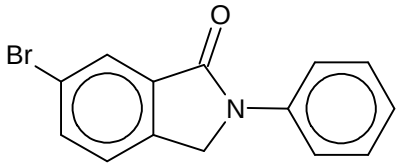
This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

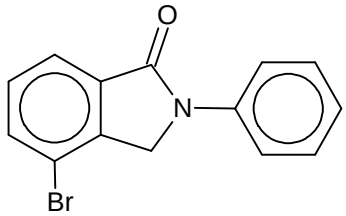
1. At 80°C the vapour pressure of pure liquid A is 520 mm Hg and that of pure liquid B is 1000 mm Hg. Liquid A and liquid B form an ideal solution. If the mixture of liquid A and liquid B boils at 80°C and 1 atm pressure, the mole fraction of A the liquid mixture is:
   (A) 0.4 (B) 0.2
   (C) 0.5 (D) 0.8

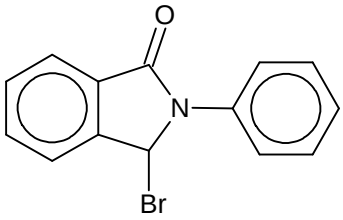
2. When 20 g of $CaCO_3$ were put into 10 litre flask and heated to 794°C, 40% of $CaCO_3$ decomposed at equilibrium. K_P for the decomposition of $CaCO_3$ is:
   (A) 0.7 atm (B) 0.3 atm
   (C) 1.05 atm (D) 2.5 atm

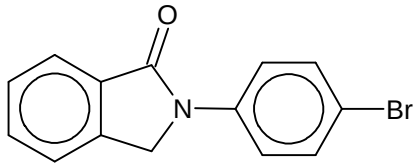
3. In the reaction given below, the major product formed is:



(A)


(B)


(C)


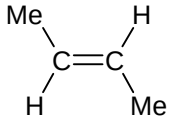
(D)


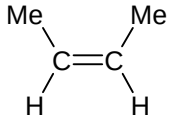
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4. In the following reactions:

$$Me-CH=CH-Me \xrightarrow[CCl_4]{Br_2} A \xrightarrow{2NaNH_2} B \xrightarrow{Na/Liq.NH_3} C$$

The product (C) is:

(A) 

(B) 

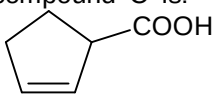
(C) $Me - CH_2 - CH_2 - Me$

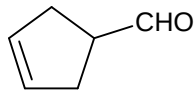
(D) $Me - C \equiv C - Me$

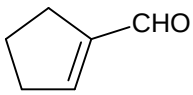
5. 50 ml of a mixture of NaOH and Na_2CO_3 solution titrated with $\frac{N}{10}$ HCl using phenolphthalein indicator required 50 ml HCl to decolourise phenolphthalein. At this stage methyl orange was added and addition of acid was continued. The second end point was reached after further addition of 10 ml of $\frac{N}{10}$ HCl. The amount of NaOH in the solution is:

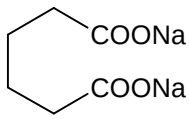
(A) 3.2 g
(B) 0.16 g
(C) 0.08 g
(D) 0.4 g

6. Cyclohexene on ozonolysis followed by reaction with zinc dust and water gives compound B. Compound B on further treatment with aqueous NaOH followed by heating yields compound C. The compound 'C' is:

(A) 

(B) 

(C) 

(D) 

7. Which among the following statement is incorrect?
(A) With excess ammonia, chlorine gives N_2 and NH_4Cl.
(B) With hot and conc. NaOH, Cl_2 gives a mixture of $NaClO_3$, NaCl and H_2O.
(C) HCl reacts with finely powdered iron to form ferric chloride.
(D) Bond dissociation energy of F – F is less than that of Cl – Cl.

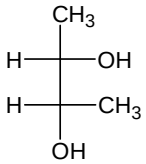
8. Which of the following is used as a chain stopping unit in the preparation of linear polymer from the hydrolysis of $(CH_3)_2SiCl_2$?
(A) $(CH_3)_3SiCl$
(B) $(CH_3)_4Si$
(C) CH_3SiCl_3
(D) $SiCl_4$

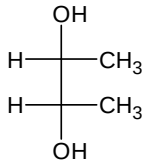
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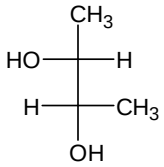
Multiple Correct Choice Type

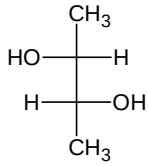
This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. Which of the following structures is/are meso-2,3-butanediol

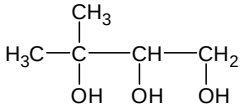
(A) 

(B) 

(C) 

(D) 

10. Which of the following compounds are formed by the oxidation of



with HIO_4

(A) $CH_3 - CHO$ (B) CH_3COCH_3

(C) $H - COOH$ (D) $HCHO$

11. For the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$; the forward reaction at constant temperature is favoured by

(A) Adding inert gas at constant volume (B) Adding inert gas at constant pressure

(C) Adding Cl_2 at constant volume (D) Increasing volume of the container

12. Which of the following statements are correct:

(A) The flocculating value of PO_4^{3-} is less than that of SO_4^{2-}.

(B) Charcoal adsorbs gases with higher critical temperature more than gases with lower critical temperature.

(C) Chemisorption is multilayered.

(D) In the manufacture of NH_3 by Haber's process, molybdenum is used as promoter for the catalyst iron.

13. In which of the complex ions the central metal ion is dsp^2 hybridized?

(A) $[Cu(NH_3)_4]^{2+}$ (B) $[Ni(CN)_4]^{2-}$

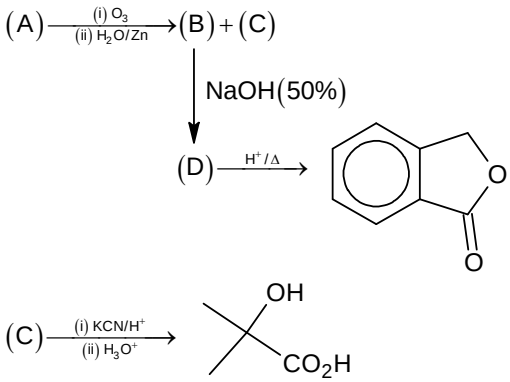
(C) $[CuCl_4]^{2-}$ (D) $[NiCl_4]^{2-}$

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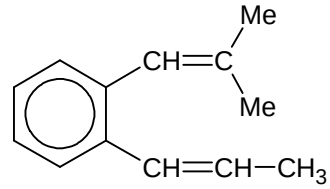
Comprehension Type

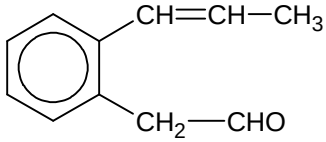
This section contains **2 paragraphs**. Based upon one of paragraph **2 multiple choice questions** and based on the other paragraphs **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of **which ONLY ONE** is correct.

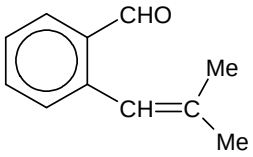
Paragraph for Question Nos. 14 to 15

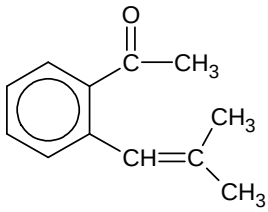


14. Structure of (A) is:

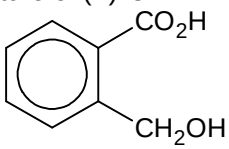
(A) 

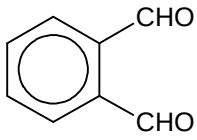
(B) 

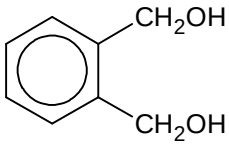
(C) 

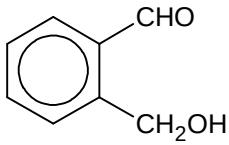
(D) 

15. Structure of (B) is:

(A) 

(B) 

(C) 

(D) 

Space for rough work

Paragraph for Question Nos. 16 to 18

A white metal sulphide (A) gives (B) gas with rotten egg smell and (C) a colourless sulphate, when treated with dil. H_2SO_4. (B) reacts with $K_2Cr_2O_7/H^+$ to form (D), a white grey element. (D) burns in oxygen to yield (E), a colourless gas. If (B) is added in (E), it gives (D) and a colourless liquid which turns anhydrous $CuSO_4$ blue. (C) gives a precipitate with NaOH which dissolves in excess of NaOH.

16. The metal sulphide (A) is:
(A) CuS (B) ZnS
(C) NiS (D) MnS

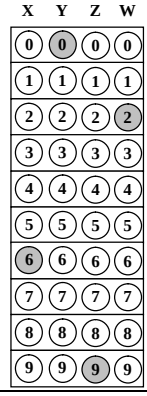
17. The compound (C) is:
(A) $CuSO_4$ (B) $MnSO_4$
(C) $ZnSO_4$ (D) $NiSO_4$

18. The colourless (E) is:
(A) H_2S (B) SO_3
(C) SO_2 (D) CO_2

SECTION – C

Integer Answer Type

This section contains **10 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. A compound AB has zinc blende structure with 'A' atoms at corners and face centre of the unit cell. If all the atoms present in one body diagonal are removed, the formula of the compound becomes A_xB_y. The sum (x + y) is:

2. Among the following, the number of compounds which undergo hydrolysis is
NCl_3, P_4O_{10}, $SiCl_4$, $BiCl_3$, XeF_4, $POCl_3$ and B_2H_6.

Space for rough work

3. The number of β-particles emitted by the conversion of $^{238}_{92}U$ to $^{206}_{82}Pb$ is

4. Total number of cyclic isomers of molecular formula C_4H_7Cl is

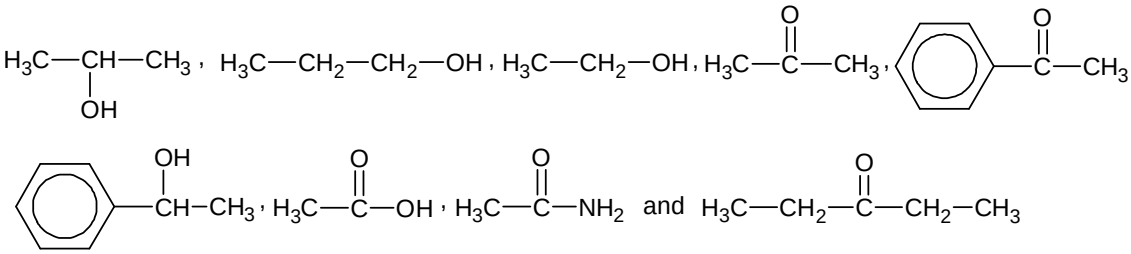
5. Total number of optically active isomers of molecular formula $C_5H_{12}O$ is

6. One mole of PCl_3 is dissolved in excess of water. Number of moles of NaOH required to neutralised this solution completely is

7. A hydrocarbon $C_xH_{12}(g)$ and $O_2(g)$ were taken in a closed flask at 400 K. When sparked they reacted to form $CO_2(g)$ and $H_2O(g)$ and no reactants left after the reaction. The pressure in the flask is increased by 20%. The value of 'x' is

8. The number of neutral oxides among the following oxides is
N_2O, NO, N_2O_3, NO_2, N_2O_5, CO, CO_2, SO_2 and CrO_3

9. Among the following the total number of compound which give positive iodoform test is:



10. The number of compounds among the following in which the sulphur has +6 oxidation state is
$H_2S_2O_5$, $H_2S_2O_4$, $H_2S_2O_6$, H_2SO_5, $H_2S_2O_8$, $H_2S_2O_7$, H_2SO_4 and H_2SO_3

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. If $|z - 1| + |z + 3| \le 8$, then the range of values of $|z - 4|$ is
   (A) (0, 7) (B) (1, 8)
   (C) [1, 9] (D) [2, 5]

2. Two circles with radii $\sqrt{2}$ cm and 1 cm meet at a point A, whose centres lie on x-axis. The distance between their centres is 2 cm. The chord AC of the larger circle cuts the smaller circle at a point B and is bisected by that point. The length of chord AC is
   (A) $\frac{\sqrt{5}}{2}$ cm (B) $\sqrt{\frac{7}{2}}$ cm
   (C) $\frac{\sqrt{3}}{2}$ cm (D) $\sqrt{\frac{5}{2}}$ cm

3. Suppose P is an interior point of a ΔABC such that the ratio $\frac{d(A, BC)}{d(P, BC)} = \frac{d(B, CA)}{d(P, CA)} = \frac{d(C, AB)}{d(P, AB)} = \frac{p_1}{p_2}$, then
   (A) P is orthocenter and $p_1 : p_2 = 4 : 1$ (B) P is orthocenter and $p_1 : p_2 = 3 : 1$
   (C) P is centroid and $p_1 : p_2 = 3 : 1$ (D) P is circumcentre and $p_1 : p_2 = 4 : 1$

4. Two triangles ABC and DEF are formed by three tangents and the corresponding three normals to a parabola $y^2 = 4ax$ respectively. The orthocenter of ΔABC has coordinates (–5, 4). Then, distance of orthocenter of ΔDEF from axis of parabola is
   (A) 5 units (B) 2 units
   (C) 4 units (D) cannot be determined

5. All the 7 digit numbers containing each of the digits 2, 3, 5, 6, 7, 8, 9 exactly once and not divisible by 5 are arranged in increasing order. The 2017th number in the list is
   (A) 6298753 (B) 6325789
   (C) 6352789 (D) none of these

Space for rough work

6. If P and Q are symmetric matrices and PQ = QP, then which of the following is/are true?
(A) $QP^{-1} = P^{-1}Q$ (B) PQ is skew symmetric
(C) $P^{-1}Q$ is symmetric (D) both (a) and (c)

7. Let W_1 and W_2 are two distinct points in an argand plane. If $p|W_1| = q|W_2|$, then the point $\frac{pW_1}{qW_2} + \frac{qW_2}{pW_1}$ is a point on the
(A) line segment [–2, 2] of the real axis (B) line segment [–2, 2] of the imaginary axis
(C) unit circle $|z| = 1$ (D) the line $\arg W = \tan^{-1} 2$

8. There are n urns $u_1, u_2, \ldots, u_n$. Each urn contains $2n+1$ balls. The ith urn contains 2i number of balls $P(u_i)$, i.e., probability of selecting i^{th} urn is proportional to (i^2+2). If we randomly select one of the urns and draw one ball and probability of ball being white be $P(A)$, then $\lim_{n\to\infty} P(A)$ is
(A) $\frac{3}{4}$ (B) $\frac{1}{4}$
(C) $\frac{2}{3}$ (D) $\frac{3}{16}$

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. All values of $x \in \left(0, \frac{\pi}{2}\right)$ such that $\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$ are
(A) $\frac{\pi}{15}$ (B) $\frac{\pi}{12}$
(C) $\frac{11\pi}{36}$ (D) $\frac{3\pi}{10}$

10. Let a, b, c be distinct complex numbers with |a| = |b| = |c| = 1 and z_1, z_2 be the roots of the equation $az^2 + bz + c = 0$ with $|z_1| = 1$. Also let P and Q represents the complex numbers z_1 and z_2 in the complex plane with $\angle POQ = \theta$ where O is the origin then
(A) $b^2 = ac$, $\theta = \frac{2\pi}{3}$ (B) $\theta = \frac{2\pi}{3}$, $PQ = \sqrt{3}$
(C) $b^2 = ac$, $PQ = 2\sqrt{3}$ (D) $2b^2 = ac$, $\theta = \frac{\pi}{3}$

Space for rough work

11. If the sum of the first 100 terms of an AP is –1 and the sum of the even terms lying in the first 100 terms is 1, then

(A) common difference is $\frac{3}{50}$ (B) first term is $-\frac{149}{50}$

(C) common difference is $\frac{7}{50}$ (D) first term is $-\frac{151}{50}$

12. A line segment is divided so that the lesser part is to the greater part is equal to the greater part is to the whole. If R is the ratio of the lesser part to the greater part, then

(A) $R^2 + R - 1 = 0$ (B) $R^2 + R^{-1} = 2$

(C) $R^{(R^2+R^{-1})} + R^{-1} = 2$ (D) $R^2 + R + 1 = 0$

13. If 'a' and 'b' are two numbers such that $a^2 + b^2 = 7$ and $a^3 + b^3 = 10$, then

(A) greatest value of $|a + b|$ is 5 (B) greatest value of $(a + b)$ is 4

(C) least value of $(a + b)$ is 1 (D) least value of $|a + b|$ is 1

Comprehension Type

This section contains **2 paragraphs**. Based upon one of paragraph **2 multiple choice questions** and based on the other paragraphs **3 multiple choice questions** have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of **which ONLY ONE** is correct.

Paragraph for Question Nos. 14 to 15

Read the following write up carefully and answer the following questions:

Dice A has four red and two white faces whereas dice B has two red and four white faces. A coin is tossed once. If it falls head, the game continues by throwing dice A, if it falls tail, dice B is to be used

14. The probability of getting a red face at any throw is

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$

(C) $\frac{1}{4}$ (D) $\frac{25}{31}$

15. If red turns up at the first 5 thrown, what is the probability that dice A is being used?

(A) $\frac{32}{33}$ (B) $\frac{30}{31}$

(C) $\frac{1}{2}$ (D) $\frac{25}{31}$

Space for rough work

Paragraph for Question Nos. 16 to 18

Read the following write up carefully and answer the following questions:
P is variable point on the line L = 0. Tangents are drawn to the circle $x^2 + y^2 = 4$ from P to touch Q and R. The parallelogram PQSR is complicated

16. If $L \equiv 2x + y = 6$, then the locus of circumcentre of ΔPQR is
(A) $2x - y = 4$ (B) $y + 2x = 3$
(C) $x - 2y = 4$ (D) $x + 2y = 3$

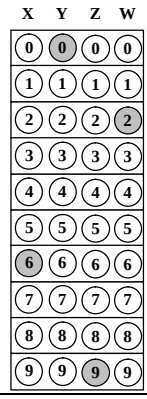
17. If $L \equiv y = 4$ then the locus of the circumcentre of ΔPQR is
(A) $y = 2$ (B) $x = 2$
(C) $y = 4$ (D) $x = -4$

18. If P = (2, 3), then the circumcentre of ΔQRS is
(A) $\left(\frac{2}{13}, \frac{7}{26}\right)$ (B) $\left(\frac{2}{13}, \frac{3}{16}\right)$
(C) $\left(\frac{4}{13}, \frac{5}{26}\right)$ (D) none of these

SECTION – C

Integer Answer Type

This section contains **10 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. Given that α, β, γ are the roots of cubic equation $x^3 - 3x^2 + 2x + \frac{2}{3} = 0$. The value of $\alpha^4 + \beta^4 + \gamma^4$ is equal to _____

Space for rough work

2. If the substitution x = cos θ reduces the differential equation $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$ to $\frac{d^2y}{d\theta^2} + Ay = 0$ then $(A^2 + 2)$ is equal to _____

3. $\lim\limits_{x\to 0^+} \frac{x}{4}\left[\frac{16}{x}\right]$ is equal to _____ (where [.] denote greatest integer function)

4. In a triangle ABC, the least values of $\Pi \operatorname{cosec}\left(\frac{A}{2}\right)$ and $\sum \operatorname{cosec}^2 A$ are p and q respectively, then (p – q) is equal to _____

5. Let $\vec{r}\times\vec{a} = \vec{b}\times\vec{a}$ and $\vec{c}:\vec{r} = 0$, where $\vec{a}\cdot\vec{c} \neq 0$, then $\vec{a}\cdot\vec{c}(\vec{r}\times\vec{b}) + \vec{b}\cdot\vec{c}(\vec{a}\times\vec{r})$ is equal to _____

6. If $g(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & x(x+1) \\ 3x(x-1) & x(x-1)(x-2) & x(x^2-1) \end{vmatrix}$, then g(200) is equal to _____

7. $13^{99} - 19^{93}$ is divisible by 162, then the remainder is _____

8. Given $(2+6+10+14+.....+2a)+(2+6+10+14+.....+2b) = (2+6+10+14+.....+2c)$, where $a+b+c = 21$ and a > 6. Then $a-b$ is equal to _____

9. Seven digit numbers are formed using digits 1, 2, 3, 4, 5, 6, 7, 8, 9 without repetition. The probability of selecting a number such that product of any 5 consecutive digits is divisible by either 5 or 7 is p. Then 12p is equal to _____

10. If a singular matrix $A = \left[a_{ij}\right]_{2\times 2}$ always commute with $B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ such that $\frac{a_{11}}{a_{12}} = \sqrt{k}$, then k is equal to _____

Space for rough work